

Dear Johns Hopkins DSAI Hiring Committee,

I am applying for the Research Software Engineer – Image Analysis & Microscopy position. I am drawn to the Data Science and AI Institute's commitment to turning project-specific solutions into sustainable scientific infrastructure. My career combines a foundation in computer science, a PhD in Neuroscience, more than 20 years of experimental biology research, and extensive experience developing scientific software. I previously served as a postdoctoral fellow and research associate in the Solomon H. Snyder Department of Neuroscience at Johns Hopkins Medicine.

My experimental career has given me substantial expertise in neuroscience, vascular biology, and cardiac physiology. My microscopy experience includes two-photon imaging of in vivo and ex vivo preparations and light-sheet and confocal imaging of fixed and cleared tissue. I have also built custom microscopy systems integrating experimental hardware with real-time acquisition and visualization software. This experience helps me translate imaging constraints into scientifically valid analysis software. I enjoy pairing collaborators' domain knowledge with my engineering experience to develop reusable, reproducible software across projects.

AcqStore and CloudScope demonstrate how I build image-analysis platforms. AcqStore is a general-purpose Python backend with public APIs for loading, organizing, and analyzing microscopy data. Plugins extend its loading, analysis, and export capabilities, while parallel execution and lazy access support batch processing and datasets too large to load in full. CloudScope uses the same backend for interactive analysis and human-in-the-loop curation, allowing researchers to inspect results against raw images and preserve review decisions. Together, these systems demonstrate my experience with the position's data-science pathway: transforming raw data into structured measurements through scalable processing pipelines and data models.

I separate computation, storage, and presentation so analytical methods can be reused across graphical applications, scripts, and notebooks. Schema-defined Zarr collections preserve images, metadata, parameters, and results, while OME-Zarr supports interoperability. Testing, continuous integration, documentation, and reproducible distribution support long-term maintenance and turn project solutions into reusable scientific infrastructure.

My broader portfolio follows the same approach. MapManager supports longitudinal tracking of neuronal structures and rapid expert review across collections containing tens of thousands of annotations, helping prevent annotation errors from obscuring small biological changes. Brightest Path implements Numba-accelerated A* tracing in n-dimensional images and provides both a documented Python package and an interactive napari interface.

I remain a hands-on developer while guiding architecture, coordinating work, and mentoring engineers. NIH and Chan Zuckerberg Initiative funding has enabled me to build multidisciplinary teams of software developers and research biologists. I also teach scientific computing and mentor trainees across computer science and engineering.

I would welcome the opportunity to bring my microscopy expertise, algorithm design, and experience building reusable scientific software to Johns Hopkins DSAI. I am excited to collaborate with faculty, scientists, and other RSEs to build open, maintainable software that advances imaging research across disciplines. Thank you for considering my application.

Sincerely,

Robert H. Cudmore, PhD